

UNIT – 1: FUNDAMENTALS OF FINANCIAL LITERACY

Introduction to Financial Literacy:

Financial literacy refers to the **knowledge, skills, and practical ability** required to manage money responsibly. It involves understanding how money works and using that understanding to make **smart financial decisions** in daily life. This includes activities such as **budgeting, saving, spending wisely, avoiding unnecessary debt, and investing for the future.**

Importance of Financial Literacy:

1. **Better Financial Decision-Making:** Financial literacy helps people compare financial products, understand interest rates, evaluate investment options, and choose what's best for their needs.
 2. **Avoiding Debt and Financial Traps:** Understanding **EMIs, loan terms, and credit card charges** helps individuals avoid high-interest loans, unnecessary borrowing, and debt cycles.
 3. **Helps Achieve Short & Long-Term Goals:** Financial literacy promotes clear planning for goals like: higher education, buying a home, medical needs or retirement savings, making them easier and more realistic.
 4. **Protection Against Scams and Fraud:** People with basic financial knowledge can identify **fake investment schemes, phishing attempts, Ponzi schemes, and misleading offers**, protecting their wealth.
 5. **Reduces Financial Stress:** Knowing how to budget, save, and plan creates a sense of security. Individuals feel more confident when dealing with unexpected expenses or economic changes.
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Advantages of Being Financially Literate:

1. **Efficient Money Management:** Financially literate individuals can plan how much to save, spend, and invest. They avoid impulse buying and focus on financial priorities.
2. **Smart Borrowing and Loan Management:** They understand the concepts of **simple interest, compound interest, loan tenure, and hidden charges**, helping them choose the right type of loan and repay it properly.
3. **Better Investment Decisions:** A financially literate person knows how to compare **risk, return, time horizon, and inflation**, enabling better investment in mutual funds, FDs, stocks, or government schemes.

4. **Awareness of Insurance and Risk Protection:** Knowing the importance of **health insurance**, **life insurance**, and **asset protection** reduces financial vulnerability.
 5. **Improved Tax Planning:** Understanding taxable income, deductions, and exemptions helps reduce tax liability and increase disposable income.
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Core Components of Financial Literacy:

These components work together to create a strong foundation for financial stability, reduced debt, disciplined saving, and confident investing.

1. Budgeting:

- Budgeting refers to the process of creating a structured plan for how income will be used. It ensures that money is allocated in a balanced way across the four major uses of money: **spending**, **saving**, **investing**, and **giving**.
- It provides financial clarity, prevents overspending, and supports long-term goals by helping individuals live within their means.

2. Investing:

- Investing involves putting money into assets with the expectation of earning returns in the future. To invest effectively, individuals must understand key investment factors such as: interest rates, inflation, diversification, risk and return.
- Learning these concepts helps people make smarter investment decisions that increase income over time and build long-term wealth.

3. Borrowing:

- Borrowing refers to taking money as a **loan**, with an agreement to repay it with interest. The fundamentals of responsible borrowing include: Interest rates, Simple interest and compound interest, Time value of money, Loan structure and repayment schedules.
- Proper borrowing decisions prevent excessive debt, reduce financial stress, and help individuals maintain control over their finances.

4. Taxation:

- Taxation is the money individuals must pay to the government on different types of income. Understanding taxes is an important part of financial literacy because it directly affects **net income**.
- Different sources of income are taxed differently, awareness of taxation enables better financial planning and improves income management.

5. Personal Financial Management: Personal financial management combines all the components of financial literacy—**budgeting, saving, borrowing, tax planning, and investing**—to create a balanced and secure financial life.

Essentials of Financial Planning

1. Setting Clear Financial Goals: Financial planning starts with defining **short-, medium-, long-term goals** like education, home purchase, or retirement. Which helps decide how much to save, how to invest, and the time period required to achieve them.

2. Analysing Income and Expenses: This step involves checking **income, expenses, debts, savings habits** to identify areas where unnecessary spending can be reduced.

3. Building an Emergency Fund: An emergency fund protects against unexpected events like illness, job loss, or urgent repairs.

4. Creating a Saving and Investment Strategy: A good financial plan includes regular saving and selecting suitable investments based on **risk level, time horizon, and expected returns**.

5. Managing Debt Wisely: Wise debt management includes borrowing only when needed and selecting loans with manageable EMIs and interest rates.

6. Reviewing and Adjusting the Plan: Financial planning must be reviewed regularly because income, responsibilities, and economic conditions change.

Basic Financial Terminology:

1. Money: Money is a recognised medium of exchange in the economy. It is an asset that can be stored and used in the form of currency, or as value.

2. Currency: Currency is the physical form of money in the form of coins and rupees. Each country typically has its own currency as a medium of exchange, issued by the central bank. In India, the Government of India (GoI) and Reserve Bank of India (RBI) are the issuers of the currency, i.e. Indian Rupees

3. Bank: A bank is a government authorised financial institution which acts as a custodian of money deposited by account holders and uses the collected funds to extend loans to individuals and businesses while charging interest on the same.

4. Account: An account is a repository of the funds held by a bank on behalf of the account holder. An account can be of various kinds, and is identified by a unique account number issued to the account holder.

5. Saving: Savings is the amount of money that is remaining from income, after the expenses are made. Investment An investment refers to an asset acquired with the objective of generating income or appreciation.

6. Internet Banking: Electronic payment system that enables customers of a bank or other financial institution to conduct a range of financial transactions through the financial institution's website.

Simple Interest (SI): Simple interest is the interest calculated only on the **initial principal amount**. It does not include interest earned on previously accumulated interest.

Formula: $SI = (P \times R \times T) / 100$

Where:

- **P** = Principal amount
- **R** = Rate of interest
- **T** = Time in years

Key Characteristics:

- Interest remains **constant** each year.
- Growth is **linear**, not exponential.
- Suitable for short-term loans or deposits.

Compound Interest (CI): Compound interest is interest earned on the **principal plus accumulated interest**. This means the interest earned each period is **added back** to the principal, causing the money to grow faster.

Formula: $A = P (1 + r/n)^{nt}$ & $CI = A - P$

Where:

- **A** = Total accumulated amount
- **P** = Principal
- **r** = Annual interest rate
- **n** = Compounding frequency per year
- **t** = Time in years

Why CI is more powerful:

- Interest is earned on both **principal** and **previous interest**.
- Money grows **exponentially**.
- Longer time periods dramatically increase returns.

Parameter	Simple Interest	Compound Interest
Definition	Simple Interest can be defined as the sum paid back for using the borrowed money over a fixed period of time.	Compound Interest can be defined as when the sum principal amount exceeds the due date for payment, along with the rate of interest for a period of time.

Formula	S.I. = $(P \times T \times R)/100$	C.I. = $P(1+R/100)^t - P$
Return Amount	The return is much lesser when compared to C.I.	The return is much higher.
Principal Amount	The principal amount is constant.	The principal amount keeps on varying during the entire borrowing period.
Growth	The growth remains quite uniform in this method.	The growth increases quite rapidly in this method.
Interest Charged	The interest charged on is for the principal amount.	The interest charged on it is for the principal and accumulated interest.

Power of Compounding:

The **power of compounding** refers to the ability of money to grow faster because interest is reinvested. The longer the money stays invested, the greater the impact of compounding. Example:

Year	Principal	Rate of return	Return earned	Maturity value
1	1,000	9%	90	1,090
2	1,090 (1000+ interest earned of 90)	9%	98	1,188
3	1,188 (1,000+interest earned 90+98)	9%	295	1295
10	2,172	9%	195	2,367*
20	5,142	9%	463	5,604
40	28,816	9%	2,593	31,409

Rule of 72:

The **Rule of 72** is a simple mathematical shortcut used to estimate how long it will take for an investment to **double in value** based on a fixed annual rate of return.

Formula: Time to Double (in years) = $72 \div \text{Interest Rate}$

Example:

If an investment earns **6%** interest per year: **$72 \div 6 = 12$ years**

This means your money will double from ₹200 to ₹400 in approximately **12 years**.

Inflation:

Inflation refers to the **continuous increase in the general price level** of goods and services over time. When inflation rises, the **purchasing power** of money falls. Inflation occurs when production costs rise, demand increases, or when there is excess money circulating in the economy.

Inflation affects everyone: households, businesses, and investors. Even if your income remains the same, rising prices reduce your ability to buy necessities, save, or invest effectively.

Example: A **Vada Pav** that cost ₹2 five years ago now costs ₹7.

There is no change in the size or quality — the rise in price is due to inflation, which increases the cost of ingredients, labor, transport, and rent.

Key Effects of Inflation:

- **Money loses value over time**
 - **Savings grow slower** if interest earned is low
 - **Future expenses become higher**
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Effects of Inflation on Investments

1. Inflation Reduces Real Returns: Real return is what remains **after subtracting inflation** from the nominal return.

Even if an investment shows growth on paper, inflation can silently reduce its effectiveness.

Example:

- Nominal return = **6%**
- Inflation rate = **5%**
- Real return = **6% – 5% = 1%**

This means the investment grew, but your purchasing power increased only slightly.

2. Inflation Makes Long-Term Goals Costlier: As prices rise year after year, long-term financial goals become more expensive. Common areas affected heavily by inflation include:

- Higher education fees
- Medical treatment and health care
- Housing and property prices
- Daily cost of living

Therefore, individuals must plan investments that beat inflation to afford these goals in the future.

3. Inflation Reduces Purchasing Power: Money saved in low-interest accounts (like basic savings accounts or cash kept at home) **loses value rapidly** during inflation.

If inflation is 7% and your savings account gives 3%, your money is effectively losing 4% value every year.

Idle money becomes weaker with time unless invested in instruments that give **inflation-adjusted or inflation-beating returns**.

4. Inflation Influences Investment Decisions: When inflation rises, investors typically shift to options that offer:

- **Higher returns**
- **Protection against rising prices**

Examples: equity mutual funds, index funds, inflation-linked bonds, real estate.

Low-yield instruments like traditional FDs may fail to keep up with inflation during high-inflation periods.

5. Inflation Creates Uncertainty in Financial Planning: When inflation is unpredictable, future expenses cannot be estimated accurately.

This affects:

- Long-term planning
- Retirement planning
- Goal-based investment strategy

Thus, estimating **future costs** with expected inflation becomes essential.

What Is GDP?

Gross Domestic Product (GDP) is the total monetary value of all **final goods and services produced within a country** during a specific period, usually one financial year. It is the most widely used measure to understand the **size, health, and growth** of a nation's economy.

GDP includes the value of **newly produced** goods and services only. It does not include resale items (like used cars or second-hand houses) because they were counted in the year they were originally produced.

Expenditure Approach to Measuring GDP

The **expenditure approach** is one of the main methods used to calculate GDP. It measures economic activity by **adding up all spending** on final goods and services made by households, businesses, governments, and foreign buyers.

GDP Formula: $GDP = C + I + G + (X - M)$, where:

Consumption (C): Household spending on goods and services—everything from groceries and rent to entertainment and healthcare.

Investment (I): Business expenditures on capital equipment (machinery, buildings) and changes in inventories. Includes new residential construction.

Government Purchases (G):Expenditures by central , state, and local governments on goods and services (e.g., defence, education, infrastructure).

Exports (X): Value of domestically produced goods and services sold abroad, adding to domestic production.

Imports (M): Value of foreign-produced goods and services purchased domestically; subtracted because they're not produced locally.

Example Calculation

Data for a hypothetical economy (in ₹ billions):

Component	Value
Consumption (C)	1,200
Investment (I)	300
Government Purchases (G)	400
Exports (X)	250
Imports (M)	150

Calculation:

$$\text{GDP} = 1,200 + 300 + 400 + (250 - 150) = \mathbf{2,000 \text{ ₹ billion}}$$